

Robust & Stochastic Control

MEAM 5170

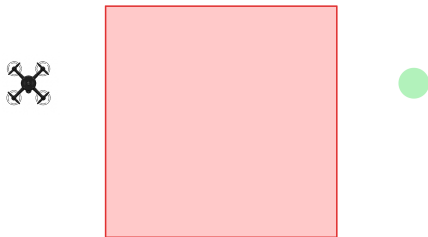
Last Time: Trajectory Optimization

Here is a general form of discrete-time trajectory optimization:

$$\begin{aligned} \min_{x_{0:T}, u_{0:T-1}} \quad & c_T(x_T) + \sum_{t=0}^{T-1} c(x_t, u_t) \\ \text{s.t.} \quad & x_{t+1} = f(x_t, u_t), \quad x_0 = x_{\text{init}}, \end{aligned}$$

A Quadrotor Example

Say we want to find a minimum length trajectory for a quadrotor to get to point A to point B:



Question: *what could go wrong?*

Defining a Stochastic Control System

Previously:

$$x_{t+1} = f(x_t, u_t), \quad x_0 = x_{\text{init}}$$

Now:

$$x_{t+1} = f(x_t, u_t, w_t), \quad x_0 \sim p(x_0),$$

where w_t is a random process.

Continuous vs Discrete

We can talk about stochastic systems in either continuous or discrete time. For simplicity, this lecture mostly focuses on discrete time.

In continuous time, you deal with **stochastic differential equations**, and have to worry about Itô's lemma, which describes how the chain rule works for SDE's.

For example, if W_t is brownian noise, it follows from Itô's lemma that:

$$\frac{d}{dt} [W_t^2] = 2W_t \frac{dW_t}{dt} + 1$$

Example: Stochastic LTI

Consider:

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad w_t \sim \mathcal{N}(0, \Sigma)$$

For some noise covariance $\Sigma = \Sigma^\top \succ 0$

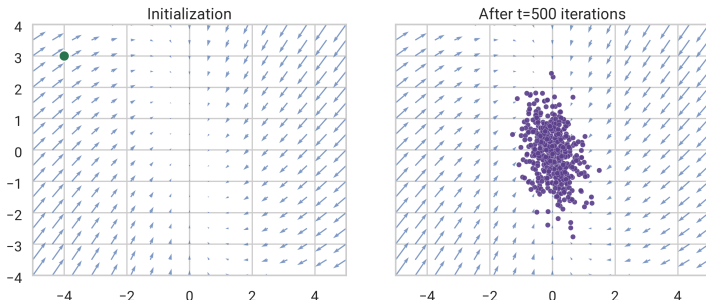
Note: if x_0 is a Gaussian and u_t is an affine function of x_t , then all x_t are Gaussian.

Let $u_t = Kx_t$ such that $\rho(A + BK) < 1$.

Question: Does the system have a fixed point at $x = 0$?

Stationary Distributions

For stochastic systems, it often makes more sense to talk about a *stationary distribution* as opposed to a stationary point.



Question: For the STLI system, if $x_0 = 0$, what happens to x_t as $t \rightarrow \infty$?

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad w_t \sim \mathcal{N}(0, \Sigma) \text{ i.i.d.}$$

Example: Manipulator Equations with Uncertainty

Recall:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = Bu$$

We might have uncertainty about physical parameters (e.g. mass, friction).
We can represent those by ξ :

$$M(q, \xi)\ddot{q} + C(q, \dot{q}, \xi)\dot{q} + g(q, \xi) = Bu$$

Then we would get the system:

$$\begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix} = \begin{bmatrix} \dot{q} \\ M(q, \xi)^{-1} (Bu - C(q, \dot{q}, \xi)\dot{q} - g(q, \xi)) \end{bmatrix}$$

where our random process, $w_t = \xi$ is a random constant (over time).

Disturbances vs Uncertainty about Parameters

Stochastic system:

$$x_{t+1} = f(x_t, u_t, w_t), \quad x_0 \sim p(x_0),$$

We might want to separate random parameters ξ that remain constant over time:

$$x_{t+1} = f(x_t, u_t, w_t, \xi), \quad x_0 \sim p(x_0)$$

Where ξ is a random variable

Note: *Random parameters can be wrapped into the state by creating an augmented state (but then your controller loses full observability)*

Observability

Our current formulation,

$$x_{t+1} = f(x_t, u_t, w_t)$$

doesn't give us uncertainty about state from partial-observability. This requires introducing observations:

$$y_t = h(x_t, u_t, v_t)$$

where v_t is a stochastic process handling measurement noise.

Note: *We now have a partially-observable markov decision process.*

Robust vs Stochastic Control

$$x_{t+1} = f(x_t, u_t, w_t), \quad x_0 \sim p(x_0),$$

Let $J(\pi; w_{0:T-1})$ be a cost function over a policy based on $w_{0:T-1}$.

Let $\pi \in \Pi$ be policy, which in the case of trajectory optimization may be parameterized by the individual control inputs $u_0, u_1, \dots, u_{T-1}$.

In general, **stochastic control** aims to optimize the expected objective:

$$\min_{\pi \in \Pi} \mathbb{E}_w J(\pi; w_{0:T-1})$$

Whereas **robust control** aims to optimize the worst-case objective:

$$\min_{\pi \in \Pi} \max_{w_{0:T-1} \in W} J(\pi; w_{0:T-1})$$

Robust and Stochastic Value Functions

We can think about value functions for a running cost $c(x, u, w)$

Stochastic:

$$V(x) = \min_u \mathbb{E}_w [c(x, u, w) + \gamma V(f(x, u, w))]$$

Robust:

$$V(x) = \min_u \max_w [c(x, u, w) + \gamma V(f(x, u, w))]$$

Question: *what is the problem with these formulations?*

Example: Domain Randomization in RL

Reinforcement learning with domain randomization tries to minimize some cost:

$$J(\theta) = \mathbb{E}_{\xi} \mathbb{E}_{\tau \sim \pi_{\theta}, \xi} \left[\sum_{t=1}^T \gamma^t c(x, u) \right]$$

where ξ are random domains (friction, etc.)

Interestingly, domain randomization can some times result in a better policy on the nominal dynamics than not doing domain randomization:

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Example: Stochastic LQR

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad w_t \sim \mathcal{N}(0, \Sigma)$$

Consider the cost $c(x, u) = x^\top Qx + u^\top Ru$ for $Q = Q^\top \succeq 0$, $R = R^\top \succ 0$. Then, our value function is:

$$V(x) = \min_u \mathbb{E}_w [x^\top Qx + u^\top Ru + \gamma V(Ax + Bu + w)]$$

Let's assume our optimal policy is $u_t = Kx_t$ and our value function is $V(x) = x^\top Sx + c$ for some $S = S^\top \succ 0$ and c . Then, we get:

$$S = Q + \gamma A^\top SA - \gamma^2 A^\top SB(R + \gamma B^\top SB)^{-1} B^\top SA$$

$$c = \frac{\gamma}{1 - \gamma} \mathbb{E}_w [w^\top Sw] = \frac{\gamma}{1 - \gamma} \text{tr}(S\Sigma)$$

Question: what happens if $\gamma = 1$?

Example: LQG

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Example: Tube Trajectory Optimization

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H_∞ Control

Consider a system with disturbances $w(t)$ and cost-relevant outputs $z(t)$:

$$\dot{x}(t) = f_\pi(x(t), w(t)), \quad z(t) = h_\pi(x(t), w(t))$$

where f_π, h_π depend on the chosen policy π .

Define a norm over a time-varying function:

$$\|x(\cdot)\|_{L_2} = \sqrt{\int_0^T x(t)^\top x(t) dt}$$

The H_∞ problem is then:

$$\min_{\pi \in \Pi} \max_{\|w\|_{L_2} \neq 0} \frac{\|z(\cdot)\|_{L_2}}{\|w(\cdot)\|_{L_2}}$$

Lyapunov Functions as Certificates in H_∞ Control

Suppose we want to build a certificate of our objective:

$$\max_{\|w\|_{L_2} \neq 0} \frac{\|z(\cdot)\|_{L_2}}{\|w(\cdot)\|_{L_2}} \leq \gamma.$$

This is equivalent to saying the following holds for all $\|w\|_{L_2}$:

$$\|z(\cdot)\|_{L_2}^2 - \gamma^2 \|w(\cdot)\|_{L_2}^2 \leq 0.$$

We can actually certify this by finding a Lyapunov function $V(x)$ satisfying:

$$\dot{V}(x) + z(t)^\top z(t) - \gamma^2 w(t)^\top w(t) \leq 0.$$

For all $w(\cdot)$.

Chance Constraints

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Other Measures of Risk: VaR and CVaR

Let J be a random variable denoting some sort of cost (e.g. the trajectory cost over a random w_t). Maybe we don't want to just focus on $\mathbb{E}J$.

The value-at-risk at level α is a quantile:

$$\text{VaR}_\alpha(J) = \min\{z : \mathbb{P}(J \leq z) \geq \alpha\}$$

The conditional value-at-risk is the expected tail cost:

$$\text{CVaR}_\alpha(J) = \mathbb{E}[J \mid J \geq \text{VaR}_\alpha(J)]$$

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(Manipulation) Funnels

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Distributionally Robust Control

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